

Exercise 2 - Theoretical question #1: solvent effects

How the solvent was included in Exercise 1

The water calculations of Exercise 1 used the conductor-like polarizable continuum model, CPCM(Water). It is an implicit solvent model: no water molecules are present. The solvent is replaced by a homogeneous, polarizable dielectric with the macroscopic constants of water, $\epsilon = 80.4$ and refractive index $n = 1.33$.

Cavity. The molecule sits in a cavity of molecular shape built from overlapping atomic spheres. By default ORCA 5 uses a van der Waals type surface with sphere radii 1.2 times the atomic van der Waals radii (C 1.70, H 1.10, O 1.52, Cl 1.75 Å); for 2-chlorophenol this gives C 2.0400, H 1.3200, O 1.8240 and Cl 2.1000 Å. The surface is discretised on Lebedev points (755 points here), and each point carries one surface charge.

Apparent surface charges. The charge distribution of the solute polarises the dielectric, and this polarisation is represented by charges q_i on the cavity surface. For a perfect conductor ($\epsilon \rightarrow \infty$) the total electrostatic potential on the surface must vanish, $V(r_i) + \sum_j V_{q,j}(r_i) = 0$, which in matrix form is $Aq = -V$: V_i is the potential of the solute (electrons and nuclei) at point i and A is the Coulomb interaction matrix between the surface charges. For a real dielectric the conductor charges are scaled down, $Aq = -f(\epsilon)V$ with $f(\epsilon) = (\epsilon - 1)/(\epsilon + x)$. CPCM uses $x = 0$, so $f = 0.9876$ for water; $x = 0.5$ would give the COSMO variant.

Gaussian charges. ORCA 5 smears each surface charge into a spherical Gaussian instead of using a point charge, so that $A_{ii} = \zeta_i (2/\pi)^{1/2}/F_i$ and $A_{ij} = \text{erf}(\zeta_{ij} r_{ij})/r_{ij}$. The switching function F_i smoothly removes charges that move inside a neighbouring sphere. With point charges, surface points appear and disappear as the atoms move, and the energy jumps; with Gaussian charges the energy and its derivatives are continuous. This is what makes the analytic gradients and Hessians usable, and both were needed for the Opt and Freq runs in water.

Self-consistent reaction field. The surface charges add a potential $V_{\text{solv}}(r) = \sum_i q_i/|r - r_i|$ (with Gaussian smearing) to the one-electron part of the Kohn–Sham operator. The charges depend on the density and the density depends on the charges, so both are updated in every SCF iteration until they are converged together. The SCF minimises the free energy $G = \langle \Psi | \hat{H}^0 | \Psi \rangle + \frac{1}{2} \langle \Psi | \hat{V} | \Psi \rangle$; the factor $\frac{1}{2}$ accounts for the work spent polarising the solvent. The orbitals, charges, dipole moment and frequencies of Exercise 1 are therefore those of the polarised, solvated molecule, which is why the dipole moment and the O–H charge separation increase in water.

What the energy contains. The solvent term printed as “CPCM Dielectric” is the electrostatic part of the solvation free energy: -0.00746 Eh ($-4.68 \text{ kcal mol}^{-1}$) with PBE and -0.00763 Eh ($-4.79 \text{ kcal mol}^{-1}$) with B3LYP at the optimised structures. Plain CPCM in ORCA 5 does not add the non-electrostatic terms (cavity formation, dispersion, repulsion); the output states that the cavity-dispersion term is not included.

Other ways to treat solvent effects in ORCA

CPCM with the COSMO scaling. ! CPCM(solvent) keeps the same model but uses $x = 0.5$ in $f(\epsilon)$.

SMD. The Solvation Model based on Density (Marenich, Cramer and Truhlar) uses the CPCM electrostatics with its own intrinsic radii and adds a non-electrostatic cavity–dispersion–solvent–structure (CDS) term, computed from atomic surface tensions and solvent descriptors (refractive index, hydrogen-bond acidity and basicity, surface tension, aromaticity, halogenicity). It is parameterised against experimental solvation free energies and is the usual choice when ΔG_{solv} itself is needed (%cpcm smd true end in ORCA 5, ! SMD(solvent) in ORCA 6).

OpenCOSMO-RS (ORCA 6). Starts from the conductor-screening charges and adds a statistical-thermodynamics treatment of the interacting surface segments. It gives solvation free energies in pure solvents and mixtures and their temperature dependence.

Implicit models for xTB. For semi-empirical GFN-xTB calculations ORCA passes the solvent to the xtb program: ALPB, ddCOSMO or CPCM-X.

Explicit and mixed solvation. Specific solute–solvent interactions, such as a water molecule hydrogen-bonded to the OH group of 2-chlorophenol, are not described by a continuum. They can be added as explicit solvent molecules, either placed by hand or automatically with the ORCA SOLVATOR (ORCA 6), and the solvated cluster can itself be embedded in CPCM (cluster–continuum model). Larger explicit solvent shells can be treated with QM/MM, using the ORCA MM module for the solvent, and molecular dynamics with explicit solvent (as in Exercise 6) samples the solvent configurations.

Excited states. For TD-DFT, CPCM is applied in linear-response form; vertical excitations are normally computed with non-equilibrium solvation, in which only the electronic polarisation of the solvent ($\epsilon_{\infty} = n^2$) follows the fast change of the density.